Geometry Mediates Excitation, Resonance, and Identifiability in Fluid-Filled Viscoelastic Shells

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Abstract

The resonant behaviour of fluid-filled viscoelastic shells arises repeatedly in biomechanics, postharvest sensing, and non-destructive evaluation, yet each application is typically treated in isolation. This paper advances a unifying thesis: *shell geometry simultaneously filters the forcing field, organises the modal spectrum, and determines the invertibility of the corresponding parameter-estimation problem.*

We formalise this thesis through four results. First, we prove that equivalent-radius (spherical) reductions of non-spherical shells are *rank-deficient* in the sense of structural identifiability: geometry and material parameters become confounded, rendering independent recovery impossible. Second, we show that asphericity lifts this rank deficiency through curvature-dependent mode splitting, providing the additional spectral information needed for unique inversion. Third, we derive the near-spherical conditioning asymptotics that govern the transition from identifiable to non-identifiable regimes, classifying shells by their curvature parameter ε. Fourth, we demonstrate that forward-model adequacy does not imply inverse-model adequacy: an equivalent-sphere model may predict resonant frequencies within engineering tolerance yet fail catastrophically when used for parameter estimation.

These four results are supported by cross-application evidence drawn from the human abdomen (infrasound resonance), the watermelon (acoustic ripeness assessment), and the urinary bladder (fill-dependent resonance), demonstrating that geometry is the hidden variable unifying apparently disparate vibroacoustic phenomena.

Keywords: fluid-filled shells; viscoelastic resonance; structural identifiability; inverse problems; asphericity; Kac problem

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